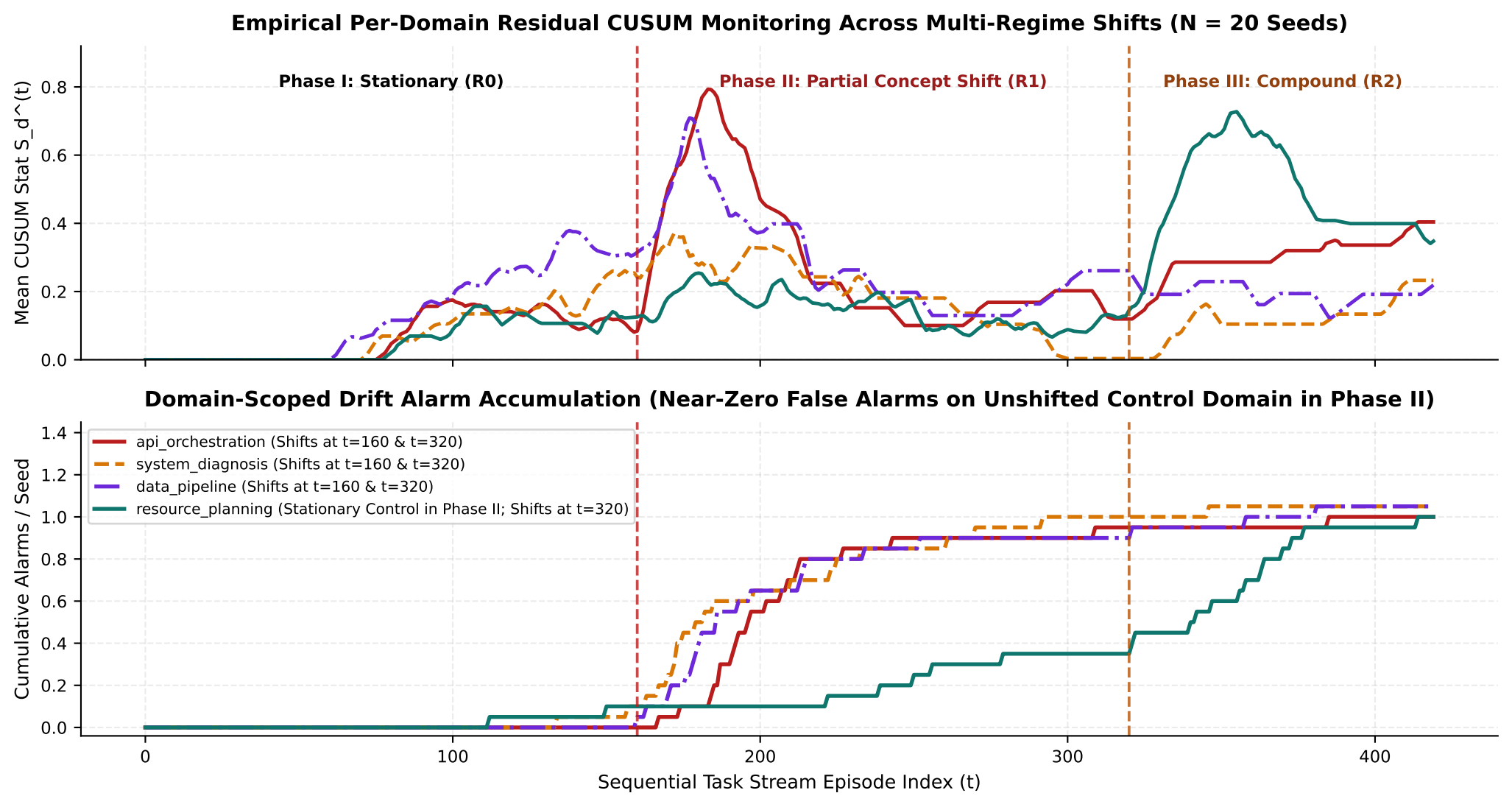
Empirical Per-Domain Residual CUSUM Monitoring Across Multi-Regime Shifts (N = 20 Seeds)
Phase I: Stationary (R0)
Phase II: Partial Concept Shift (R1)
Phase III: Compound (R2)
Mean CUSUM Stat S_d^(t)
0.0
0.2
0.4
0.6
0.8
Domain-Scoped Drift Alarm Accumulation (Near-Zero False Alarms on Unshifted Control Domain in Phase II)
api_orchestration (Shifts at t=160 & t=320)
system_diagnosis (Shifts at t=160 & t=320)
data_pipeline (Shifts at t=160 & t=320)
resource_planning (Stationary Control in Phase II; Shifts at t=320)
Cumulative Alarms / Seed
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
100
200
300
400
Sequential Task Stream Episode Index (t)